Phase cycling in CP-FTMW

What it removes, what it does not, and how to choose the pattern.

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1. Why there are several chirps in one buffer

A gas pulse lasts around a millisecond but a chirp/FID cycle only a few microseconds, so many cycles fit inside one pulse. The scope, not the AWG, sets the throughput — about 32 acquisitions per second whatever you do, some 21 ms of re-arm each. Putting K cycles in one buffer and firing ONE marker means that re-arm is paid once per gas pulse instead of once per FID.

Once the cycles are there anyway, giving each one a different chirp phase costs nothing and buys a filter. That is phase cycling.

2. How it works

Cycle k transmits its chirp with a programmed phase phi_k. Anything that responds to the chirp carries that phase back. The K sub-FIDs are then recombined after de-rotating each one by the phase it was sent with:

$$S = \frac{1}{K}\sum_{k=0}^{K-1} e^{-i\varphi_k} z_k$$

The molecular signal picks up exactly phi_k, so every term lands at zero phase and the K add coherently. Anything that ignores the chirp has the same phase in every cycle, so de-rotating spreads it around the circle instead and it sums to

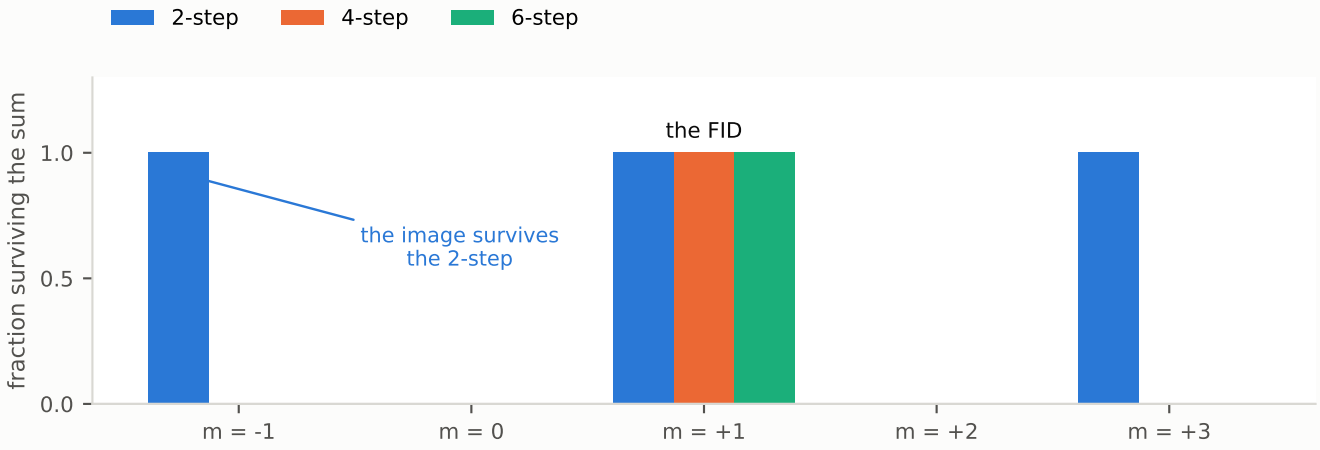
$$\sum_{k} e^{-i\varphi_k}$$

which is the whole condition. The pattern works if and only if those phasors close. 0/180 closes. 0/90/180/270 closes. 0/120/240 closes. What does NOT close is a partial repetition — K = 4 running a 3-step pattern gives 0, 120, 240, 0, whose phasors sum to 1, not 0, and nothing cancels at all.

> **K must be a whole number of repetitions of the pattern. Not approximately — a partial cycle gives no rejection whatsoever, not reduced rejection.**

3. Which signals it filters out

Write a component's response to a chirp of phase phi as exp(i·m·phi). The integer m is its coherence order: the FID is m = +1, and anything indifferent to the chirp is m = 0. Substituting into the sum above, an N-step cycle passes order m only when m − 1 is a multiple of N.




```
order   meaning                                  2-step   4-step   6-step
m = +1   THE FID -- the molecular signal           1.00     1.00     1.00
m = +0   chirp breakthrough, detector ringing      0.00     0.00     0.00
m = -1   image: I/Q imbalance, folded mirror       1.00     0.00     0.00
m = +2   second-order response                     0.00     0.00     0.00
m = +3   third-order response                      1.00     0.00     0.00
```

Both patterns kill m = 0, which is the large one in practice: chirp breakthrough into the receiver, detector ringing, DC offset and digital pickup all sit there. Two steps are enough for that.

The difference that matters is m = −1, the image — a receiver I/Q imbalance, or a mirror sideband folding into the IF. With an even number of steps m = −1 satisfies m − 1 = −2, a multiple of 2, so the 2-step passes it at full strength. It adds coherently with the signal and does not look like an artifact. The 4-step rejects it because −2 is not a multiple of 4.

> **Use the 2-step on a bare bench, where only breakthrough and ringing exist. Use the 4-step once there is a receiver, because that is when an image term becomes possible.**

4. Measured, not just derived

Measuring the artifacts that happen to be present proves only that the cycling does no harm — on a bare AWG-to-scope path they are already 52 dB down. So two artifacts of known size and known order were injected into the same buffer as the chirp, placed outside its sweep so each could be measured by an exact complex projection: a constant-phase tone at 2.64 GHz (m = 0) and a tone carrying −phi_k at 1.84 GHz (m = −1), both at 30% of the chirp.

```
pattern        chirp m=+1     m=0 artifact     m=-1 image
no cycling         4.00x          -0.0 dB        -0.0 dB
2-step             4.00x         -51.8 dB        -0.0 dB
4-step             4.00x         -48.3 dB       -52.7 dB
```

4 cycles, 50 GS/s, 16 hardware averages, 2026-10-26. The chirp adds 4.00x in every case. The 2-step row is the one to read: its m = −1 column is identical to no cycling at all. The floor around −50 dB is the bench, not the method — it matches the residual measured with nothing injected.

5. What it does NOT remove

If the same molecules are hit by more than one chirp — which happens whenever the cycle period is shorter than a molecule's transit across the beam — consecutive chirps produce a

Hahn echo carrying phase 2·phi_k − phi_(k−1). De-rotating by phi_k leaves phi_k − phi_(k−1), which for a uniform N-step pattern is the constant 2π/N. It therefore adds coherently in every cycle, and no number of steps changes that:

```
  pattern        echo surviving
  2-step              1.00
  4-step              1.00
  6-step              1.00
  4-step, order scrambled      0.53
```

Scrambling the order within a balanced set holds m = 0 and the image at exactly zero while roughly halving the echo, which is a cheap option if it turns out to matter.

There is a structural protection, though. With chirps at uniform spacing T the echo from chirps k−1 and k refocuses at 2t_k − t_(k−1) = t_(k+1), exactly when the next chirp fires, and the same holds for any pair. In a rigidly uniform train every two-pulse echo lands under a later chirp and never inside a detect window.

> **Keep the cycle spacing exactly uniform. It is what keeps echoes out of the detect windows, and phase cycling will not clean them up if they get in.**

6. Choosing K

K is the window the sample is present for, over the cycle period, rounded DOWN to a whole number of patterns. A 1 ms gas pulse is no constraint — some 88 cycles fit at a 11.1 µs period and you want all of them, a cycle being ~2000x cheaper than a new pulse. For laser ablation the floor is what matters: 4 cycles for the 4-step, 2 for the 2-step.

```
  30 us window, chirp 1 us:   usable K (multiple of 4)
  dead     FID 0.5us   FID 1us   FID 2us   FID 4us   FID 6us
  0.2 us         16        12         8         4         4
  0.5 us         12        12         8         4         4
  1   us         12         8         4         4        --
  2   us          8         4         4         4        --
  5   us          4         4        --        --        --
```

Collecting past the FID decay adds noise and no signal, so the best detect window IS the decay time, not the longest that fits. Doppler width scales with frequency, so a 350 GHz FID decays far faster than the same transition at 10 GHz and the cycle is dominated by chirp plus dead time — on an ablation window the dead time alone can swing K by five, a factor of two in sensitivity.

7. Where this lives in the code

```
awg.py        send_waveforms(timing={'n_cycles': K}) plus a per-
              channel 'phases' list; _build_channel lays the K
              copies and removes the free-running NCO advance.
phasecycle.py split_cycles() cuts at exact times, combine()
              returns BOTH sums so artifacts stay measured.
gui.py        the Phase cycling block; tests/ the offline check.
```

Two traps live there: the NCO advance must be removed per cycle, and cycles cut at exact times, not rounded samples — a 0.44-sample slip is 28 degrees at 2.24 GHz and cost a 4-cycle sum 15%.